

School of Computer and Information Technology

Beaconhouse National University

Software Engineering for CS

Fall 2025 - Project Proposal

Section C

Group Members

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Company/ Product Name

ClassSync AI is an intelligent timetable generation system designed to automate and optimize the complex scheduling process for universities. It uses a custom Genetic Algorithm to efficiently allocate resources like classrooms and instructors, significantly minimizing conflicts and maximizing resource utilization.

System Overview

ClassSync AI will be designed to replace traditional, manual timetabling methods which are notoriously time-consuming, prone to error, and result in suboptimal use of university resources. The system will solve the critical problem of scheduling inefficiency by providing a fast, reliable, and optimized solution.

The system will be built around a core set of modules designed for maximum impact:

- **Configuration & Data Input Module:** An intuitive interface for administrators to upload institutional data (courses, instructors, room details) and, most importantly, to define specific organizational rules and constraints (e.g., instructor availability, room capacity, course dependencies).
- **Intelligent Scheduling Engine:** At the heart of the system is an AI-powered optimization engine, inspired by genetic algorithms. This engine will process all inputs and constraints to generate thousands of potential timetable solutions, intelligently selecting the one that is mathematically proven to be the most optimal and conflict-free.
- **Real-time Conflict & Validation System:** A robust system to check and penalize violations of both hard constraints (e.g., no instructor double-booking) and soft constraints (e.g., preferred time slots, VC breaks).
- **Output Generation Module:** The system will generate clear, easy-to-read timetables that can be viewed by instructor, by room, or by instructor.

By developing this system, our goal is to provide a tool that dramatically saves administrative time, reduces scheduling conflicts to near-zero, and ensures the best possible use of university facilities. The steps to successful development will follow four key steps:

1. **Discovery & Requirements Finalization:** We will work closely with your team to document every specific rule, constraint, and desired outcome for your institution.
2. **System Design & Prototyping:** We will design the system architecture and provide interactive prototypes of the user interface for your feedback.
3. **Agile Development & Testing:** The system will be built in iterative cycles, allowing for continuous testing and feedback to ensure the final product perfectly meets your needs.
4. **Deployment, Training & Support:** We will deploy the finished software, provide comprehensive training for your administrative staff, and offer ongoing support.

Return on Investment

The primary benefit for the university is a massive gain in operational efficiency. The system reduces scheduling time from 40+ hours of manual work to under 10 minutes, freeing up valuable administrative resources. This directly translates to cost savings and allows staff to focus on other critical tasks. The investment is justified by the immediate and tangible improvements in accuracy and resource management.

As a product, our business model is a Software-as-a-Service (SaaS) platform targeting educational institutions.

- **Initial Phase Strategy:** We will pilot ClassSync AI within the SCIT department at Beaconhouse National University to create a strong case study. We will offer the service at a reduced cost to early adopters to gather feedback and build a client base. Initial marketing will focus on direct outreach to local universities and presentations at academic technology conferences.
- **5-Year Growth Plan:**
 - **Year 1:** Achieve full adoption within BNU and secure contracts with 3-5 other Pakistani universities. Focus on refining the core product based on user feedback.
 - **Year 2:** Expand our client base nationally. Begin development of planned features like cloud integration for real-time collaboration.
 - **Year 3:** Introduce tiered subscription plans based on institution size. Begin exploring international markets, particularly in regions with similar educational structures.
 - **Year 4:** Launch the cloud-based version of ClassSync AI. Integrate advanced features like Natural Language Processing (NLP) to allow administrators to input constraints conversationally.
 - **Year 5:** Establish ClassSync AI as a leading timetable optimization tool in the region. Form strategic partnerships with larger EdTech companies to integrate our solution into their university management suites. The financial return is projected to be substantial, driven by recurring subscription revenue and a growing customer base.